

Keeping the Ratio: Proportional Reasoning in Everyday Contexts

PhET Proportion Playground · PhET

Years 8-10

~30 minutes

Mathematics · Number, Ratios & Financial Maths

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/proportion-playground>

Curriculum links: AC9M8M07 · AC9M9M05 · AC9M10M05

Learning intentions

- I can find a missing quantity that keeps a given ratio constant.
- I can use ratio reasoning to solve practical scaling problems.
- I can explain my proportional reasoning using words like 'ratio', 'scale factor' and 'rate'.

Getting started

Open PhET Proportion Playground and choose one of the everyday contexts (for example, the necklace beads, paint mixing, pool balls or apples). Each puzzle shows you a starting ratio and asks you to find a missing amount that keeps the ratio the same. Try the first puzzle in a context before reading on, just to see how the checking feature works.

Tasks

1. In your chosen context, write down the starting ratio the puzzle gives you (for example, '2 blue beads to 3 red beads').

2. Work out the missing quantity needed to keep this ratio the same when the total amount changes. Show your working before checking your answer in the simulation.

Working:

3. Was your answer correct? If not, describe what mistake you made and correct your working.

4. Complete four more puzzles in this context (or switch to a new one). Record your working amount and whether the simulation confirmed you were correct.

Given amount	Your calculated missing amount	Correct? (Y/N)

5. Choose the paint-mixing context if available. Explain, using ratio language, why doubling every ingredient in a paint mixture keeps the colour the same, but changing only one ingredient does not.

6. Switch to a new context. Find a puzzle where the ratio involves three quantities instead of two (if available), or create your own by scaling a two-quantity ratio up by a chosen scale factor. Show your calculation.

Working:

7. A recipe uses a ratio of 2 cups of flour to 5 cups of water. If you wanted to make a batch using 15 cups of water, explain step-by-step (using ideas from this simulation) how you would work out how much flour to use.

8. Explain the difference between a 'ratio' and a 'rate', using one example from this simulation and one example from everyday life (e.g. speed, price per kilogram).

Extension (optional)

A juice company changes its recipe from 'ratio of 3 parts concentrate to 5 parts water' to 'ratio of 4 parts concentrate to 7 parts water'. Use proportional reasoning to explain whether the new juice will taste stronger, weaker, or the same, and justify your answer with a calculation.
